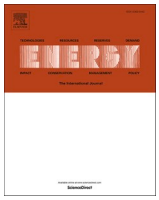
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Energy consumption modeling and optimization of an eVTOL aircraft: Integrating weight, motor, and battery dynamics

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ABSTRACT

Electric vertical takeoff and landing (eVTOL) aircraft, as a frontier technology in urban air mobility (UAM), have garnered significant attention from both academia and industry in recent years. This study addresses the high energy efficiency design challenge for eVTOL aircraft by proposing a multi-disciplinary design optimization (MDO) framework. Weight, motor efficiency, and electrochemical-aging-thermal coupled model of the battery were developed and integrated to construct a comprehensive whole-aircraft energy consumption analysis model. Using optimization algorithms, design parameters were optimized for various mission scenarios. Results demonstrate that the proposed optimization achieves an 11.44 % reduction in total energy consumption and a 15.81 % increase in maximum take-off mass compared to the baseline. Furthermore, the study identifies those key parameters, including cruise range, cruise height, and hover time, significantly influence the aircraft's energy consumption. This research not only provides theoretical foundations and design guidelines for developing commercially viable eVTOL systems with higher energy efficiency and longer battery life but also lays crucial theoretical groundwork for advancing the practical implementation and commercial application of eVTOL technology.

1. Introduction

Electric vertical takeoff and landing (eVTOL) aircraft, also known as air taxis, represent one of the most innovative technologies for urban air mobility (UAM) and have garnered significant attention from both industry and academia in recent years [1]. These aircraft offer a transformative approach to addressing urban traffic congestion, driven by the increasing global proliferation of automobiles and the growing strain on ground transportation systems. With its benefits like no runway and environmental and green protection, eVTOL aircraft is a cutting-edge approach to air travel that is predicted to grow into a significant low-altitude economy carrier. From proof of concept to initial commercialization, eVTOL technology is currently in a crucial stage. Around 500 concepts for electric vertical takeoff and landing aircraft have been unveiled since 2016 [2,3]. Numerous technology companies, as well as traditional aviation manufacturers, are actively exploring commercialization paths for eVTOL aircraft. Aircraft manufacturers such as Airbus, Lilium, and Kitty Hawk have been actively involved in manufacturing air taxis, and companies such as Bell, Embraer, Hyundai, Rolls-Royce, and Toyota have recently begun to venture into the air taxi market [4].

Currently, the main research directions concerning eVTOL aircraft are in the areas of overall dimensional design and selection [5,6], aircraft battery-related design [7–9], flight path planning [10,11], and aircraft control system design [12]. In the context of overall vehicle sizing and selection, Ugwueze et al. introduced a robust sizing methodology with a mixture of dichotomous, fixed-point, and Newton-Raphson methods, which was used to investigate the comparative performance of the wingless and powered-lift eVTOL aircraft types at different mission lengths [13]. An et al. used several methods such as constraints analysis methods for eVTOL multi-mode unmanned aerial vehicle (UAV), regression modeling for electric propulsion system selection, selection methods for electric propulsion and hydrogen fuel cell systems, and transition analysis methods to obtain UAV geometry, mass, and power data, to develop a hydrogen fuel cell and battery-powered eVTOL UAV for a selection method [14]. Wang et al. proposed a bi-layer optimization method for eVTOL electric propulsion systems, dividing the design process into two layers to maximize energy

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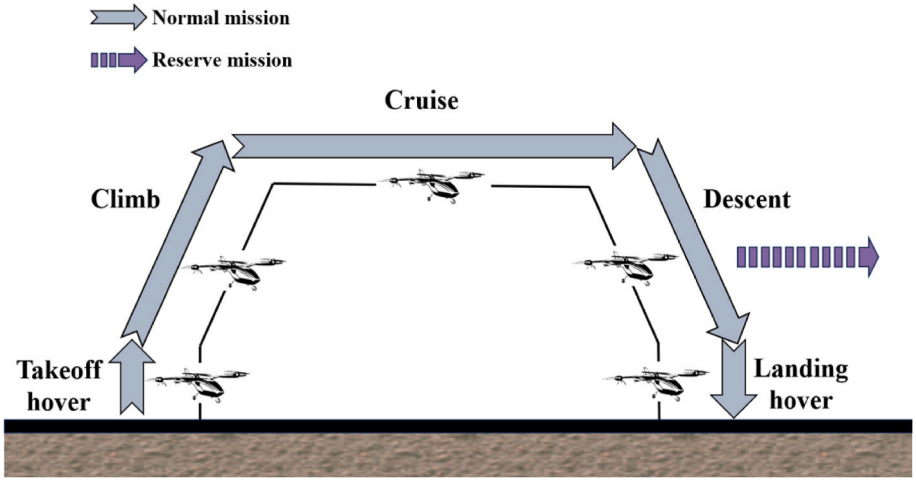


Fig. 1. Mission diagram of eVTOL aircraft.

Table 1
Summary of normal mission.

Parameter	Value
Takeoff velocity	2.5 m/s
Hover height	15 m
Hover time	60 s
Climb velocity	2 m/s
Cruise height	300 m
Cruise range	150 km

efficiency and minimize weight. Applied to a tilt-wing eVTOL aircraft, the method reduces mass by 11.66 % and offers a versatile framework for various configurations and missions [15]. In the field of aircraft battery design, Dixit et al. simulated the initial takeoff step of an eVTOL aircraft powered by lithium-ion batteries and performed extensive electrochemical testing to assess the long-term stability of lithium-ion batteries under high-strain conditions [16]. The results show the potential for battery failure under the high-rate conditions required by the eVTOL aircraft. Singh et al. applied the algorithm to fast charging through a new technique called partial incremental capacity (PIC) [17]. They conducted individual cell and supercell experiments before integrating the PIC method into the algorithm. Finally, the fault detection capability of the improved algorithm was verified using an actual eVTOL battery module. As a result, the updated algorithm can perform fault diagnosis while fast charging and is well-suited for implementation in eVTOL aircraft. In aircraft path planning, Wang et al. proposed an optimization-based maneuverability assessment method as an alternative to the pilot experiment approach. The method uses trajectory optimization to determine the achievable performance of a specific mission, which enables efficient maneuverability assessment and provides model-based analysis of eVTOL aircraft design quality [18]. Park et al. performed trajectory optimization for takeoff and landing of UAM considering the energy efficiency and safety [19]. The longitudinal model of single-seater eVTOL aircraft is used. The battery state-of-charge, estimated by the coulomb counting method, is set as the objective function and the vortex ring state avoidance constraint is applied to secure the safety. The numerical optimization is conducted using the direct collocation method, and the effects of the objective function and constraint are analyzed. In aircraft control system design, Saetti et al. developed an automatic flight control system (AFCS) for the eVTOL aircraft that provides stability and desired roll, pitch, yaw, and tilt axis response characteristics over a speed range from hover to 80 kt [20]. They also used the AFCS to perform simulations to analyze autorotation performance after a complete loss of power. Yilmaz et al. proposed a nonlinear programming formula for control assignment optimization based on nonlinear control mapping and used a sparse nonlinear optimizer to solve the problem [21]. The solutions obtained from different formulas, such as the minimization of force and torque residuals and the minimization of control force, are studied. In addition, these solutions are compared with those obtained by using the formula based on linear control mapping. While there are abundant studies on eVTOL aircraft, few studies offer a systematic analysis of their energy consumption mechanisms during flight.

Motivated by the above remarks, this work attempts to fill this research gap and makes the following contributions: (1) The coupling effects of energy consumption in eVTOL aircraft were systematically investigated through a multidisciplinary design optimization (MDO) framework. (2) The motor efficiency model and the electrochemical–aging–thermal coupling mechanisms within the battery were thoroughly investigated. These analyses established a theoretical foundation for the design of energy-efficient eVTOL aircraft and provided critical guidance for optimizing component performance. (3) An MDO-based total energy consumption model was developed for eVTOL aircraft, which was subsequently integrated with practical operational requirements to derive efficient design strategies. This integration enables balanced optimization of energy efficiency and mission-specific performance parameters.

The remainder of the work is framed as follows: The eVTOL aircraft problem is described in Section 2. The energy consumption of eVTOL aircraft is modeled in Section 3 and it is optimized in Section 4. Conclusions are provided in Section 5.

2. Problem formulation

2.1. Mission description

The normal mission of the eVTOL aircraft (Fig. 1) is divided into five phases: takeoff hover phase, climb phase, cruise phase, descent phase, and landing hover phase [13,18]. To cope with emergencies, it is also necessary to set a reserve mission, which consumes about 20 % of the normal mission.

The mission profile for the aircraft is defined as follows: during the normal mission, the aircraft takes off at a velocity of 2.5 m/s, ascends vertically to a hover height of 15 m above ground level, and hovers at this height for 60 s. Subsequently, the aircraft ascends at a velocity of 2 m/s to reach the designated cruise height of 300 m. Upon reaching the cruise height, the aircraft must fly a distance of 150 km at the designated cruise velocity, followed by a descent to the hover height at a reduced velocity. The aircraft then hovers at the hover height for an additional 60 s before landing safely [13,19,20]. The normal mission profile

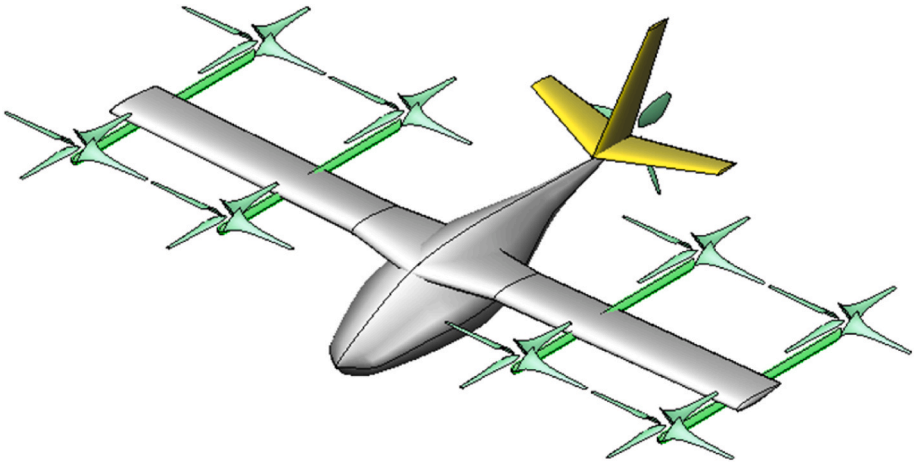


Fig. 2. Schematic diagram of the research object.

Table 2
Initial parameter setting.

Parameter	Value
Number of passengers (Pilot included)	4
Payload	400 kg
Propeller efficiency	0.8
ESC efficiency	0.85
Battery energy density	300 Wh/kg
Motor power density	6 kW/kg

described in this study is summarized in Table 1.

2.2. Basic configurations

Referring to the existing eVTOL aircraft configurations [6,21,22], it is assumed that the research object in this paper is the aircraft shown in Fig. 2, which has efficient vertical takeoff and landing capability and smooth cruise performance. The aircraft is a tilt-rotor aircraft with nine propellers, eight of which are mounted on the wing beams at the upper part of the wing, and the rest of the propellers are mounted on the tail of the fuselage.

During the takeoff and hover phase, the eight propellers located on the crossbeam collectively generate the necessary vertical thrust to enable the aircraft to ascend and maintain a stationary hover. As the

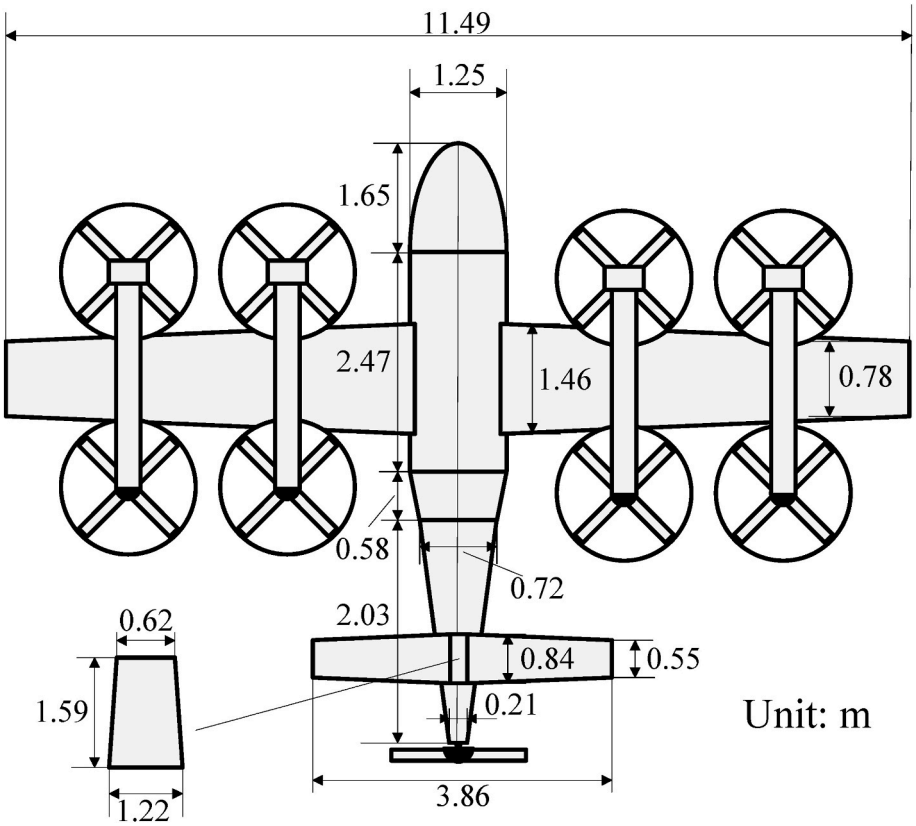


Fig. 3. Sketch of aircraft geometric dimensions.

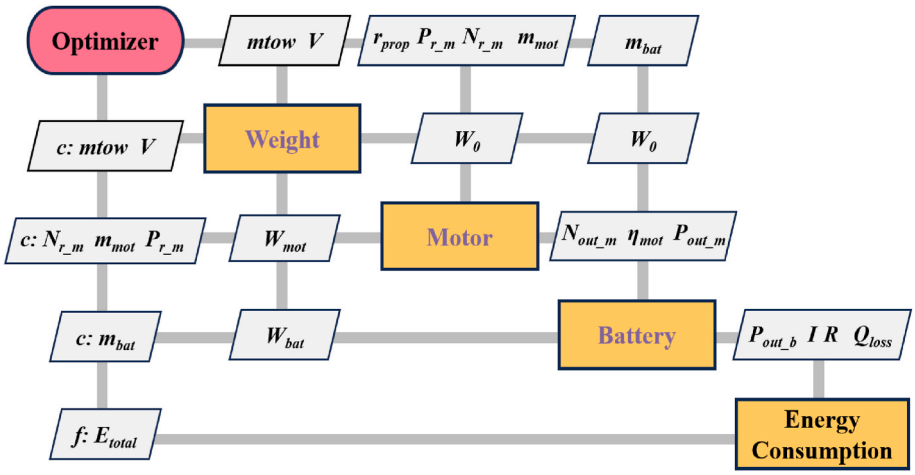


Fig. 4. Schematic of the MDO framework, showing the coupling between weight, motor, and battery subsystems.

aircraft transitions to the climb phase, these propellers continue to provide upward thrust to counteract gravitational forces and ensure a smooth ascent to the designated cruise height. Concurrently, the tail propeller supplies horizontal thrust to accelerate the aircraft until it achieves the desired cruise velocity. Upon entering the cruise phase, the four forward propellers are tilted at a 90-degree angle, redirecting their thrust from vertical to horizontal, thereby facilitating efficient horizontal flight. Finally, during the descent and hover landing phase, the crossbeam propellers adjust their thrust output to decelerate the aircraft, ensuring a smooth touchdown and maintaining hover readiness for safety and stability.

2.3. Initial parameter setting

Some related parameters of the eVTOL aircraft will be used in the subsequent optimization and calculation. The initial parameters will be set in this section (Table 2). It is assumed that the aircraft can take four people including the pilot. The tolerable payload is 400 kg. The propeller efficiency is 0.8. The efficiency of the electronic speed controller (ESC) for controlling the motor speed is 0.85. The energy density of the battery can reach 300 Wh/kg. The power density of the motor is 6 kW/kg [3,23–26].

In addition, knowing the geometric dimensions of the aircraft is essential when calculating the weight in Section 3.2. Based on previous studies [27], a reasonable approximation of the geometric dimensions of the aircraft shown in Fig. 2 has been conducted, and Fig. 3 presents its geometric dimensions.

2.4. Assumptions

Energy consumption modeling for eVTOL aircraft during actual flight is highly complex, and the computational solving of such models presents significant challenges. This study represents an initial investigation into aircraft energy consumption. Therefore, simplifications must be made to the energy consumption model before its development, in order to reduce computational complexity. The following assumptions are made.

(1) Uncertainties during the flight process are disregarded.
(2) Dynamic control behavior, wind effects, and transitional aerodynamic phenomena are excluded.
(3) The power consumption remains constant within each flight phase.
(4) The flight trajectory is assumed to be fixed, with negligible additional energy consumption due to turns or path deviations.
(5) All calculations involving air density, temperature lapse rates, and altitude are performed under standard atmospheric conditions, assuming zero wind speed.
(6) Environmental conditions such as temperature, pressure, and humidity are maintained at constant levels.
(7) The influence of coupling factors outside the three disciplines (weight, motor, and battery) on energy consumption is neglected.

3. Energy consumption modeling

3.1. MDO architecture

MDO is an innovative design methodology for addressing complex system optimization challenges. The fundamental principle of the MDO approach lies in achieving global optimum designs through the integration and synergistic interactions among interdependent disciplines [28,29]. When applied to aircraft conceptual and preliminary design, MDO enables a comprehensive understanding of the coupled characteristics inherent in diverse aerospace disciplines relevant to aircraft development, which contributes to enhanced aircraft performance and reduced design cycles.

Given the complexity of eVTOL aircraft design, the application of MDO is particularly advantageous for optimizing energy consumption. This paper aims to conduct MDO on the eVTOL aircraft to reduce its energy consumption. The MDO framework is illustrated in Fig. 4.

As shown in Fig. 4, the energy consumption of an eVTOL aircraft is closely related to its battery output power (P_{out_b}), discharge current (I), internal resistance (R), and capacity loss (Q_{loss}). A complex coupling relationship exists between the battery, motor, and weight. Specifically, the weight (W_0), impacts the thrust during takeoff hover as well as the lift during cruise, thereby affecting the output power of the motor and battery (P_{out_m}, P_{out_b}). The battery supplies power according to the motor's required output power, rotational speed (N_{out_m}), and efficiency (η_{mot}), thus influencing its output power. Additionally, the power that the motor and battery need to provide affects their weight (W_{mot}, W_{bat}), which in turn impacts the eVTOL aircraft's weight.

To comprehensively understand the total energy consumption of eVTOL aircrafts throughout their flight missions and the coupling relationships among disciplines, subsequent sections will elaborate on the weight discipline (Section 3.2), motor discipline (Section 3.3), and battery discipline (Section 3.4) within a multidisciplinary framework. Based on these analyses, a total energy consumption model will be established (Section 3.5).

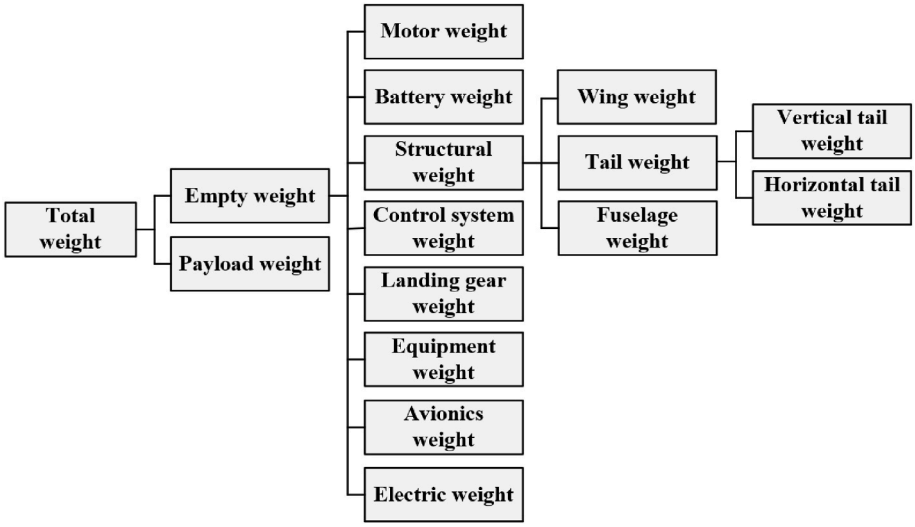


Fig. 5. Total weight composition of the eVTOL aircraft.

3.2. Weight

This section employs a statistical estimation method to conduct a detailed weight assessment of each aircraft component. Rooted in a classic framework from aerospace engineering, this method ensures high accuracy and applicability. Before initiating the estimation process, several prerequisites must be established: first, the aircraft's geometric dimensions have been determined through preliminary design processes (specific parameters are outlined in Section 2.3); second, historical data from similar aircraft are referenced to obtain a reasonable initial estimate of weight parameters. Incorporating these elements, and utilizing the provided geometric dimensions and initial weight parameters, the weight of each aircraft component is individually assessed, culminating in a precise total weight value.

The total weight typically comprises the combined weight of multiple components [30], as illustrated in Fig. 5. In this section, the methodology for estimating the weight of each component will be thoroughly discussed.

As shown in Fig. 5, the total weight W_0 of the eVTOL aircraft consists of the empty weight W_{em} and the payload weight W_{pa}:

$$W_0 = W_{em} + W_{pa} \tag{1}$$

The payload weight is generally the weight of passengers and cargo, in this case 400 × 9.8 N.

The empty weight consists of motor weight W_{mot}, battery weight W_{bat}, structure weight W_{str}, control system weight W_{con}, landing gear weight W_{lg}, equipment weight W_{eq}, avionics weight W_{av}, and electric weight W_{el}:

$$W_{em} = W_{mot} + W_{bat} + W_{str} + W_{con} + W_{lg} + W_{eq} + W_{av} + W_{el} \tag{2}$$

where $W_{mot} = 9.8 \cdot \mathrm{m_{mot}}$, $W_{bat} = 9.8 \cdot \mathrm{m_{bat}}$ (m_{mot}, m_{bat} are the mass of motor and battery).

Detailed computational processes for determining W_{str}, W_{con}, W_{lg}, W_{eq}, W_{av}, and W_{el} are presented in the Appendix.

As demonstrated in Section 3.1, W_{bat} and W_{mot} constitute coupled state variables that create interdisciplinary dependencies among three disciplines (weight, motor, and battery). To decouple the multidisciplinary constraints, the fundamental mass variables (m_{bat} and m_{mot}) are strategically elevated to independent design variables in the optimization framework, thereby enabling partitioned discipline-specific analyses.

3.3. Motor

3.3.1. Efficiency model

The motor serves as a critical component of the onboard energy system, with its internal energy losses significantly influencing the overall energy consumption of the eVTOL aircraft, and the motor's operational efficiency directly impacts energy utilization.

However, existing literature frequently employs fixed efficiency values for estimating motor energy consumption. Motor efficiency is closely linked to its operational conditions. During different flight phases, the motor's output power and speed undergo significant changes, leading to variations in efficiency. Consequently, this section will carry out a detailed analysis of the motor's efficiency characteristics under various operating conditions, aiming to reveal its internal energy loss patterns. This analysis will provide a more accurate basis for establishing an energy consumption model for eVTOL aircraft.

The motor efficiency η_{mot} is calculated by Eq. (3) [31]:

$$\eta_{mot} = \frac{P_{out_m}}{P_{in_m}} = \frac{P_{out_m}}{P_{out_m} + P_{loss_m}} \tag{3}$$

where P_{out_m} is the output power. P_{in_m} is the input power. P_{loss_m} is the loss of power.

The power loss of a motor encompasses several components, including copper loss P_{loss_Cu}, eddy current loss P_{loss_e}, hysteresis loss P_{loss_h}, mechanical loss P_{loss_ma}, and additional loss P_{loss_a}. P_{loss_Cu} arises from the electrical resistance of conductors when current flows through them. P_{loss_e} is caused by the changing magnetic flux in the core, which induces circulating currents. P_{loss_h} results from the magnetic hysteresis of ferromagnetic materials in the motor. P_{loss_ma} is attributed to friction and vibration between motor components often due to rotor imbalance or bearing wear. P_{loss_a} is primarily associated with leakage and harmonic losses generated by the stator or rotor due to the working current [32–36]. Accordingly, the total power loss P_{loss_m} can be represented as shown in Eq. (4):

$$P_{loss_m} = P_{loss_Cu} + P_{loss_e} + P_{loss_h} + P_{loss_ma} + P_{loss_a} \tag{4}$$

Meanwhile, the various power losses are related to the total losses at the peak power point. The total loss at peak power point $P_{p_loss_m}$ is calculated by Eq. (5):

$$P_{p_loss_m} = \frac{2}{3} \cdot h \cdot t \cdot A \tag{5}$$

where h represents the heat dissipation coefficient. t represents the

Table 3
Parameters used to calculate density and temperature.

Parameter	Value
ρ_0	1.225 kg/m^3
T_0	288.15 K
g	9.8 m/s^2
K_T	−0.0065 K/m
R	287.05 m^2/ks^2

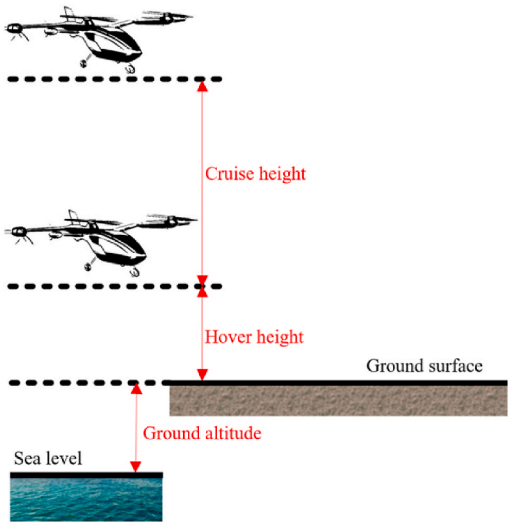


Fig. 6. Schematic diagram of the relationship among hover height, cruise height, and ground altitude.

temperature rise limit. A represents the surface area of the iron core. When the motor is at peak power, $P_{p_loss_Cu} = 0.59{\cdot}P_{p_loss_m}$, $P_{p_loss_e} = 0.11{\cdot}P_{p_loss_m}$, $P_{p_loss_h} = 0.11{\cdot}P_{p_loss_m}$. Then the power loss expression is obtained as follows [37]:

$$P_{loss_Cu} = \begin{cases} [(N_{r_m}/N_{out_m}){\cdot}(P_{out_m}/P_{max_m})]^2{\cdot}P_{p_loss_Cu} & (N_{out_m} \le N_{r_m}) \\ (P_{out_m}/P_{max_m})^2{\cdot}P_{p_loss_Cu} & (N_{out_m} > N_{r_m}) \end{cases} \tag{6}$$

$$P_{loss_e} = \begin{cases} (N_{out_m}/N_{r_m})^2{\cdot}P_{p_loss_e} & (N_{out_m} \le N_{r_m}) \\ P_{p_loss_e} & (N_{out_m} > N_{r_m}) \end{cases} \tag{7}$$

$$P_{loss_h} = \begin{cases} (N_{out_m}/N_{r_m}){\cdot}P_{p_loss_h} & (N_{out_m} \le N_{r_m}) \\ (N_{m-r}/N_{m-m})^{0.6}{\cdot}P_{p_loss_h} & (N_{out_m} > N_{r_m}) \end{cases} \tag{8}$$

$$P_{loss_ma} = 1.11 \times 10^{-5}{\cdot}N_{r_m}^{0.7}{\cdot}P_{r_m} \tag{9}$$

$$P_{loss_a} = 0.064{\cdot}\left(P_{loss_Cu} + P_{loss_e} + P_{loss_h} + P_{loss_ma}\right) \tag{10}$$

where $P_{\max_m}$ indicates the peak power. N_{r_m} indicates the rated speed. P_{r_m} indicates the rated power. N_{out_m} denotes the output speed. The resulting expression for η_{mot} in both cases is obtained [31]:

When $N_{out_m} < N_{r_m}$, the motor efficiency is calculated by Eq. (11):

$$\eta_{mot} = \frac{P_{out_m}}{P_{out_m} + \left(\begin{array}{l} (N_{r_m}{\cdot}P_{out_m}/N_{out_m}{\cdot}P_{max_m})^2{\cdot}P_{p_loss_Cu} \\ +(N_{out_m}/N_{r_m})^2{\cdot}P_{p_loss_e} + (N_{out_m}/N_{r_m}){\cdot}P_{p_loss_h} \\ +1.11 \times 10^{-5}{\cdot}N_{r_m}^{0.7}{\cdot}P_{r_m} \end{array} \right) \times 1.064} \tag{11}$$

When $N_{out_m} \ge N_{r_m}$, the motor efficiency is calculated by Eq. (12):

$$\eta_{mot} = \frac{P_{out_m}}{P_{out_m} + \left(\begin{array}{l} (P_{out_m}/P_{max_m})^2{\cdot}P_{p_loss_Cu} + P_{p_loss_e} \\ +(N_{r_m}/N_{out_m})^{0.6}{\cdot}P_{p_loss_h} + 1.11 \times 10^{-5}{\cdot}N_{r_m}^{0.7}{\cdot}P_{r_m} \end{array} \right) \times 1.064} \tag{12}$$

Using Eqs. (11) and (12), η_{mot} correlates with P_{out_m}, $P_{\max_m}$, P_{r_m}, N_{out_m}, and N_{r_m}. In this study, the nine motors installed on the eVTOL aircraft share identical hardware configurations; thus, $P_{\max_m}$, P_{r_m}, and N_{out_m} remain constant for each motor throughout the analysis. However, due to variations in P_{out_m} and N_{out_m} during different flight phases, the η_{mot} of each motor differs across these stages. Consequently, the following analysis in this section systematically examines the P_{out_m} and N_{out_m} across flight phases, with a detailed analysis of P_{out_m} presented in Section 3.3.2 and N_{out_m} analyzed in Section 3.3.3.

3.3.2. Output power model

In this study, the motor and propeller are integrated directly without incorporating a gearbox. Consequently, the motor's output power is determined by the propeller's output power P_{out_p} and its efficiencyη_{pro}, as shown in Equation (13).

$$P_{out_m} = \frac{P_{out_p}}{\eta_{pro}} \tag{13}$$

The propeller's output power at each flight phase is discussed in detail below.

(1) Takeoff hover

Each of the eight propellers is equipped with four blades, providing upward thrust to the aircraft during takeoff hover. During this phase, the required thrust is equal to the sum of the aircraft's weight. Based on this thrust and using the propeller radius and other parameters, the total propeller output power during the takeoff hover phase can be calculated [5]:

$$P_{pro,toh} = W_0{\cdot}\sqrt{\frac{T_{pro,toh}}{2{\cdot}A_{pro}{\cdot}k_{pro}{\cdot}\rho_1}} \tag{14}$$

where $T_{pro,toh}$ is the thrust generated by a single propeller, defined as Eq. (15). A_{pro} represents the area of the propeller disk, which is calculated using Eq. (16). k_{pro} denotes the correction factor for the propeller disk and taken as 0.9 in this analysis. ρ_1 corresponds to the air density at the hover altitude. When the flight altitude is below 11,000 m, the air density is determined based on Eq. (17).

$$T_{pr\ o,toh} = \frac{W_0}{n_{pro,toh}} \tag{15}$$

where $n_{pro,toh}$ represents the number of motors generating upward thrust, which equals 8 in this study.

$$A_{pro} = \pi{\cdot}r_{prop}^2 \tag{16}$$

where r_{prop} is the propeller radius.

$$\begin{cases} \rho_i = \rho_0{\cdot}\left(\dfrac{T_i}{T_0}\right)^{\frac{-g}{K_T{\cdot}R}-1} \\ T_i = T_0 - 0.0065{\cdot}h_i \end{cases} \quad (i = 1, 2) \tag{17}$$

where ρ_0 is the air density at sea level. h_i is altitude. T_i is the temperature at an altitude h_i. T_0 is the sea level temperature. g is the acceleration of gravity. K_T is the temperature gradient below the top of the troposphere. R is the air gas constant. The values of the above parameters are shown in Table 3 [38].

The relationship between the aircraft's hover height, cruise height,

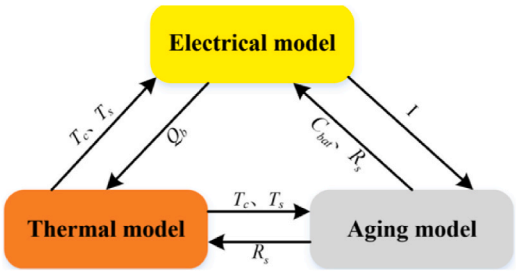


Fig. 7. The coupled electrical-thermal-aging battery model.

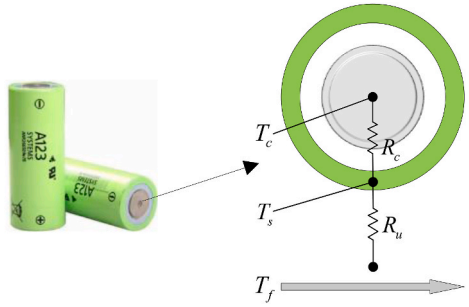


Fig. 8. Schematic diagram of dual-state thermal model.

and ground altitude is shown in Fig. 6.

(2) Climb

The output power of the propeller in the climb phase $P_{pro,cb}$ is determined by the power ratio, which is governed by the aircraft's takeoff velocity and its relative vertical velocity V_v. The power ratio between the climb phase and the takeoff hover phase is [13]:

$$\frac{P_{pro,cb}}{P_{pro,toh}} = \frac{V_v}{2 \cdot V_{toh}} + \sqrt{\left(\frac{V_v}{2 \cdot V_{toh}}\right)^2 + 1} \tag{18}$$

where V_{toh} is the takeoff velocity. When the aircraft is in the climb phase, V_v can be approximated as the climb velocity V_{cb}.

Therefore, the output power of the propeller is:

$$P_{pro,cb} = P_{pro,toh} \cdot \left(\frac{V_v}{2 \cdot V_{toh}} + \sqrt{\left(\frac{V_v}{2 \cdot V_{toh}}\right)^2 + 1}\right) \tag{19}$$

(3) Cruise

During the cruise phase, the aircraft generates the required horizontal thrust through its five propellers. Assuming the aircraft flies at a constant cruise velocity V, the horizontal and vertical forces acting on the aircraft are in equilibrium at this stage. Specifically, in the horizontal direction, the airflow resistance equals the total thrust produced by the five propellers. In the vertical direction, the lift force equals the aircraft's weight. Thus, the output power of the propellers can be expressed as:

$$P_{pro,cru} = D \cdot V \tag{20}$$

where D is the air resistance of the aircraft, which is calculated by Eq. (21) [39].

$$D = \frac{1}{2} \cdot \rho_2 \cdot S_{wi} \cdot V^2 \cdot C_D \tag{21}$$

where ρ_2 is the air density at cruise altitude, which is calculated by Eq. (17). C_D is the resistance coefficient, which is calculated by Eq. (22) [13].

$$C_D = C_{D0} + \frac{C_L^2}{\pi \cdot AR_w \cdot e} \tag{22}$$

where C_{D0} denotes the zero-lift drag coefficient, which is assumed to be 0.012 in this analysis. C_L represents the lift coefficient, adopted as 0.5 in this study. e is the Oswald efficiency coefficient, which is calculated by Eq. (23).

$$e = 1.78 \times \left(1 - 0.045 \cdot AR_w^{0.68}\right) - 0.64 \tag{23}$$

(4) Other phases

When the aircraft is in the low-speed descent and landing hover phases, the propeller's output power is similar to that during the takeoff hover phase. It is assumed that the output power during these phases is equal to that during the takeoff hover phase.

3.3.3. Output speed model

The output speed of the motor is the same as that of the propeller. The thrust generated by the propeller T_{pro} at the output speed N_{out_pro} is [40]:

$$T_{pro} = C_T \cdot \rho_i \cdot \left(\frac{N_{out_pro}}{60}\right)^2 \cdot (2 \cdot r_{prop})^4 \tag{24}$$

where C_T refers to the coefficient of thrust, here taken as 1.5.

Then,

$$N_{out_pro} = 60 \cdot \sqrt{\frac{T_{pro}}{C_T \cdot \rho_i \cdot (2 \cdot r_{prop})^4}} \tag{25}$$

In the takeoff hover phase, the thrust provided by a single propeller is calculated by Eq. (15).

In the cruise phase, the thrust generated by a single propeller $T_{pro,cru}$ is:

$$T_{pro,cru} = \frac{D}{n_{pro,cru}} \tag{26}$$

where $n_{pro,cru}$ is the number of rotors providing horizontal thrust in the cruise phase, and its value is 5.

In other flight phases, the propeller's output speed is assumed to remain consistent with that during takeoff hover, as these phases have a similar demand for vertical loads.

3.4. Battery

The total energy consumption of an eVTOL aircraft during a normal mission is closely related to the internal energy losses within its battery. Currently, there is a scarcity of studies that thoroughly investigate the internal energy loss mechanisms in eVTOL battery systems. Notably, when the battery has a substantial mass, these internal losses significantly impact the overall energy consumption. Therefore, this section employs a coupled electro-thermal-aging battery model [41,42] (Fig. 7) to study the internal energy loss characteristics of the battery. This model dynamically simulates the electrochemical, thermal, and aging characteristics of a single cell within the battery during full charge and discharge cycles using the MATLAB Simulink environment, thereby obtaining parameters that characterize the energy losses in the battery. The relevant parameters in the model are described in the following subsections.

The single cell used in this study is the A123LiFePO$_4$ cylindrical cell, with a mass of 76 g, a rated capacity of 2.5 Ah, and an internal resistance of 10 mΩ [43].

Table 4
Thermal model parameters.

Parameter	Value
C_c	4.5 J/K
C_s	62.7 J/K
R_c	1.94 K/W
R_u	3.4 K/W

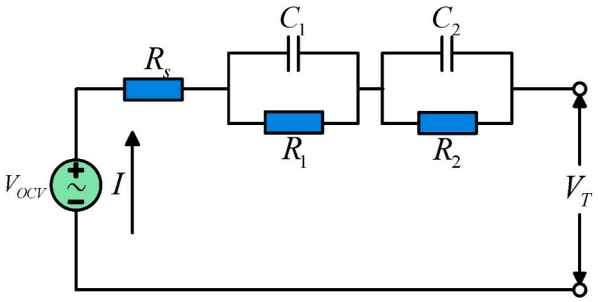


Fig. 9. The second-order resistance-capacitance equivalent circuit model.

Table 5
Aging model parameters.

Parameter	Value
α	2896.6 (SOC<45 %) 2694.5 (SOC≥45 %)
β	7411.2 (SOC<45 %) 6022.2 (SOC≥45 %)
E	31,500 J/mol
R_g	8.314 J/(mol·K)
z	0.57

3.4.1. Thermal model

During a full charge-discharge cycle, battery performance is affected by temperature variations, which subsequently influence internal parameters. The dual-state thermal model [44] (Fig. 8) is utilized to simulate battery thermal characteristics.

The radial temperature distribution of the battery is:

$$\begin{cases} C_c \cdot dT_c = Q + \dfrac{T_s - T_c}{R_c} \\ C_s \cdot dT_s = \dfrac{T_f - T_s}{R_u} - \dfrac{T_s - T_c}{R_c} \end{cases} \tag{27}$$

where T_c and T_s are the core temperature and surface temperature of the battery. dT_c and dT_s represent the rate of change of core temperature and surface temperature. T_f denotes the coolant temperature. C_c is the total heat capacity of the core. C_s is the total heat capacity of the surface. R_c is the equivalent conduction resistance between the magnetic core and the surface. R_u is the equivalent conduction resistance between the battery surface and the coolant, which depends on the nature of the coolant. The values of thermal parameters are shown in Table 4 [45]. Q represents the rate of heat generation, which is defined by the energy dissipated in the electrode overpotential and the Joule heat. Entropy heat production is not considered due to its small effect on the total heat production. Q is calculated by Eq. (28):

$$Q = I \cdot (V_{ocv} - V_{out}) \tag{28}$$

where I is the current. V_{ocv} is the open circuit voltage. V_{out} is the output voltage. These parameters will be discussed in the battery electrical model.

3.4.2. Electrical model

The electrical model is used to simulate the response electrical characteristics (voltage, current, etc.) of the battery.

The SOC of the battery is calculated by Eq. (29):

$$SOC = \frac{\int Idt}{3600 \cdot C_{bat}} \tag{29}$$

where C_{bat} is the battery capacity.

The average battery temperature is:

$$T_{mea} = \frac{1}{2} \cdot (T_s + T_c) \tag{30}$$

The second-order resistance-capacitance equivalent circuit model is adopted in this study [8] (Fig. 9). V_{ocv} has a nonlinear relationship with the state of charge. R_s stands for ohmic resistance. R_1C_1 and R_2C_2 are two RC pairs, which are affected by the average battery temperature and SOC [45].

To accurately analyze the parameters in the electrical model, it is essential to establish a rigorous understanding of the battery's behavior throughout a complete charging and discharging cycle. A comprehensive charging and discharging strategy is employed, which is divided into three sequential stages: a standard constant current (CC) charging stage, followed by a standard constant voltage (CV) charging stage, and concludes with a constant current (CC) discharging stage.

Stage 1 Standard CC charging

The charging current at this stage is input and remains constant. V_{out} is output:

$$V_{out} = V_{ocv} - V_1 - V_2 - I \cdot R_S \tag{31}$$

where V_1 and V_2 represent the voltages in the charge transfer process and the battery diffusion process. They are calculated by Eq. (33).

The rate of change of V_1 and V_2 is:

$$\begin{cases} dV_1 = -\dfrac{V_1}{R_1 \cdot C_1} + \dfrac{I}{C_1} \\ dV_2 = -\dfrac{V_2}{R_2 \cdot C_2} + \dfrac{I}{C_2} \end{cases} \tag{32}$$

Then,

$$\begin{cases} V_1 = I \cdot R_1 - D_1 \cdot \exp\left(-\dfrac{t}{R_1 \cdot C_1}\right) \\ V_2 = I \cdot R_2 - D_2 \cdot \exp\left(-\dfrac{t}{R_2 \cdot C_2}\right) \end{cases} \tag{33}$$

where D_1 and D_2 are constants. t is the charging time.

Stage 2 Standard CV charging

At this stage, input remains constant. I is output:

$$\begin{aligned} I &= \frac{1}{R_s} \cdot (V_{ocv} - V_1 - V_2 - V_{out}) \\ &= \frac{1}{R_s} \cdot (V_{ocv} - i_1 \cdot R_1 - i_2 \cdot R_2 - V_{out}) \end{aligned} \tag{34}$$

where i_1 is the current flowing through R_1 and i_2 is the current flowing through R_2.

Stage 3 CC discharging

Batteries are often discharged using CC discharging. The CC discharging model is the same as the standard CC charging and will not be repeated here.

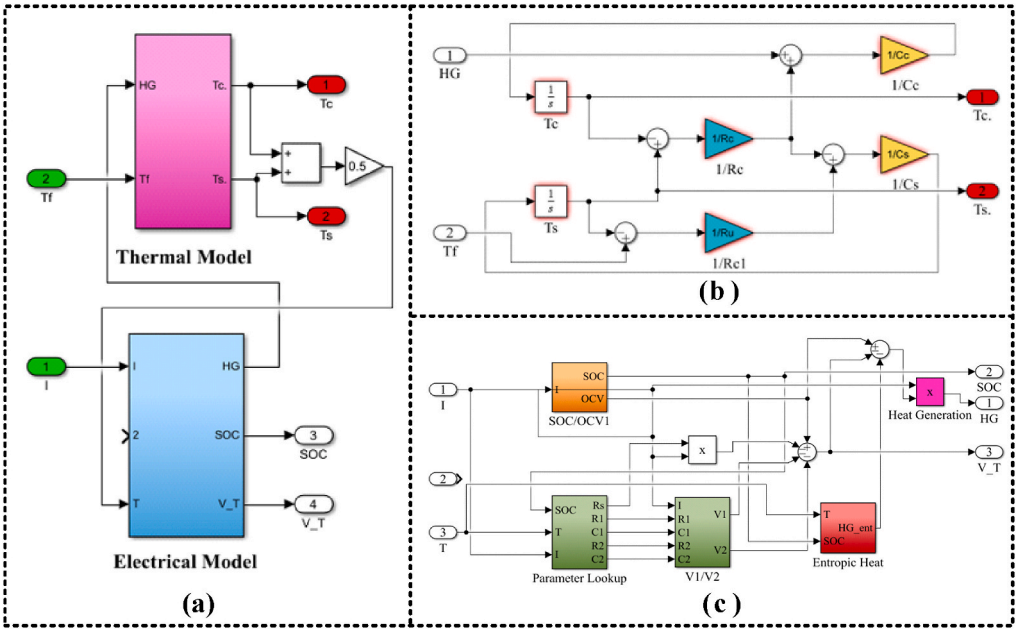


Fig. 10. Battery simulation model (a), thermal-electrical coupling model (b), and thermal simulation model (c).

3.4.3. Aging model

Assuming the eVTOL aircraft is charged after each completed normal mission, the battery will undergo frequent charge-discharge cycles due to the high annual flight volume. These cycles result in continuous degradation of battery capacity over time, which may impact the aircraft's long-term performance and battery lifespan. To ensure the battery system meets the aircraft's performance and lifespan requirements, an aging model has been incorporated into the study. The aging model is often used to describe the capacity decay rate of a battery [43]:

$$\Delta Q_{loss} = (\alpha \cdot SOC + \beta) \cdot \exp\left(\frac{-E + \eta_c \cdot I_c}{R_g \cdot (273 + T_{mea})}\right) \cdot Ah^z \tag{35}$$

where α and β are constant values, which are related to the magnitude of the *SOC*. E is the activation energy. R_g is the universal gas constant. Ah is the cumulative charge throughput, the value of which is equal to the battery capacity multiplied by the number of charge/discharge cycles. z is a constant value. The values of these parameters are shown in Table 5 [41,43].

The simulation models obtained by MATLAB Simulink are shown in Fig. 10. Through the simulation model, we can get the changes in battery parameters during charging and discharging (Fig. 11), and these parameters can be used to calculate the internal energy consumption and capacity loss of the battery.

3.5. Total energy consumption model

The total energy consumption of an eVTOL aircraft during a normal mission is the sum of the energy consumed in each flight phase. The total energy consumption model is presented in Eq. (36):

$$E_{total} = E_{takeoffHover} + E_{climb} + E_{cruise} + E_{descent} + E_{landingHover} \tag{36}$$

The energy consumption of the aircraft in each flight phase E_i is determined by the output power $P_{out_b,i}$ and internal energy consumption $E_{loss_b,i}$ of the battery [46]:

$$E_i = \int_{t_i}^{t_{i+1}} P_{out_b,i} dt + E_{loss_b,i} \tag{37}$$

The battery, as a component of the aircraft propulsion system, directly has its output power P_{out_b} determined by the system's output power. The propulsion system in this study comprises four key parts: battery, ESC, motor, and propeller [47]. P_{out_b} is transmitted through the ESC, motor, and propeller, transforming into mechanical power. During this conversion, each component inevitably incurs power losses. The output power transfer path of the battery is illustrated in Fig. 12.

The power losses within the ESC, motor, and propeller are expressed in terms of their respective efficiencies (η_{ESC}, η_{mot}, η_{pro}). η_{mot} is discussed in Section 3.3.1. Therefore, $P_{out_b,i}$ is calculated by Eq. (38):

$$P_{out_b,i} = \frac{P_{out_p,i}}{\eta_{ESC} \cdot \eta_{mot} \cdot \eta_{pro}} \tag{38}$$

where $P_{out_p,i}$ is the output power of the propeller in each flight phase, which has been discussed in Section 3.3.2.

$E_{loss_b,i}$ is calculated by Eq. (39):

$$E_{loss_b,i} = n_{cell} \cdot Q_{h,i} \tag{39}$$

where n_{cell} is the number of single cells within the battery. $Q_{h,i}$ is the heat generated by the single cell in each flight phase.

Each single cell weighs 76 g. Excluding the weight of the battery's wiring and packaging, n_{cell} is calculated by Eq. (39):

$$n_{cell} = \frac{1000 \cdot m_{bat}}{76} \tag{40}$$

The energy in the battery is mainly lost in the form of heat. $Q_{h,i}$ is calculated by Eq. (41) [40]:

$$Q_{h,i} = I^2 \cdot (R_s + R_1 + R_2) \cdot t_i \tag{41}$$

where t_i is the flight time of the aircraft at each flight phase.

Assuming that the output power of the battery is constant in each flight phase, i.e., the power does not vary with time. Eq. (37) can be rewritten as:

$$E_i = \left(P_{out_b,i} + Q_{h,i}\right) \cdot t_i \tag{42}$$

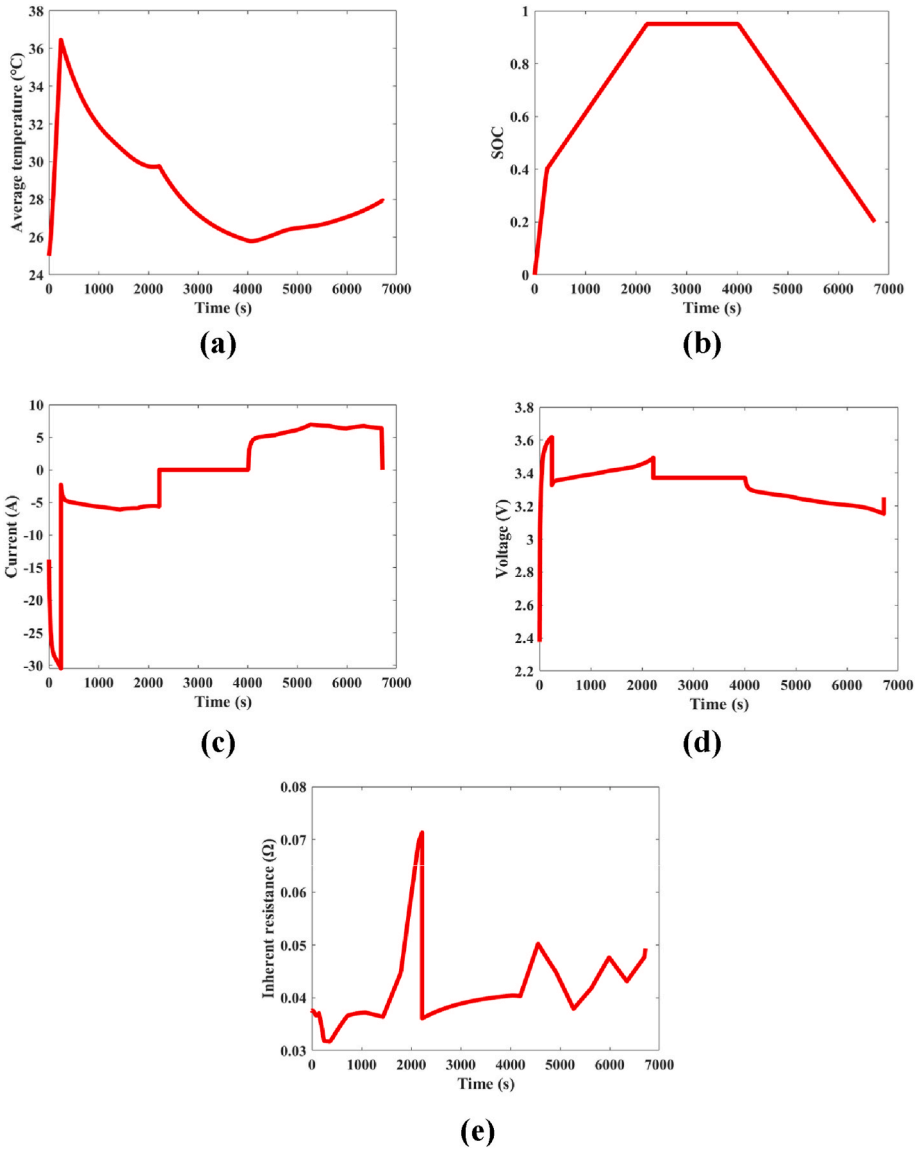


Fig. 11. Variation of average temperature (a), SOC (b), current (c), voltage (d), and inherent resistance (e).

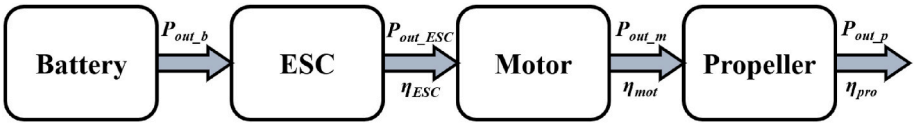


Fig. 12. Output power transfer path of the battery.

4. Energy consumption optimization

4.1. Design variables

To optimize the energy consumption of the eVTOL aircraft, the parameters that affect energy consumption are selected as design variables in this study. In Section 3, an energy consumption model is established, and the analysis reveals several critical factors: Rotor radius r_{prop} affects motor output power. Cruise velocity V alters motor power demand during the cruise. Maximum takeoff mass $mtom$ impacts total weight and energy distribution. Motor mass m_{mot} and battery mass m_{bat} jointly influence weight distribution. Motor rated power P_{r_m} and rated speed N_{r_m} determine operational characteristics and efficiency. Ultimately, these variables directly influence energy consumption across flight phases. Therefore, r_{prop}, V, $mtom$, m_{mot}, m_{bat}, P_{r_m}, and N_{r_m} are selected as design variables in this study:

$$\mathbf{x} = \left[r_{prop}, V, mtom, m_{bat}, m_{mot}, P_{r_m}, N_{r_m}\right] \tag{43}$$

4.2. Constraints

During flight, the design variables must be constrained by upper and lower bounds due to multiple constraints such as force, weight, and motor efficiency:

$$\mathbf{x}_{\min} \leq \mathbf{x} \leq \mathbf{x}_{\max} \tag{44}$$

where $\mathbf{x}_{\min}$ and $\mathbf{x}_{\max}$ are the upper and lower bounds of design variables.

In addition, the aircraft must also meet the following constraints.

Table 6
Comparison of the design parameters optimization results.

Parameter	Unit	Value range	Initial value	Baseline	Optima scheme
r_{prop}	m	[0.3,2]	1	1.33	0.97
V	m/s	[40,80]	60	59.02	62.94
m_{bat}	kg	[500,1100]	700	825.43	708.51
m_{mot}	kg	[10,50]	30	10.30	12.24
$mtom$	kg	[1000,2500]	1500	1971.49	1614.29
P_{r_m}	kW	[10,50]	30	N/A	30.17
N_{r_m}	rev/min	[2000,5000]	3500	N/A	2000

(1) The total energy provided by the battery is dictated by its mass and energy density, respectively. To meet the requirements of a flight mission, the supplied energy must exceed the total energy reserve required for the mission, which is set to 1.2 times the energy consumption of a normal mission:

$$1.2 \cdot E_{total} < \frac{m_{bat} \cdot \rho_{bat}}{1000} \tag{45}$$

(2) The power that the motor can provide is determined by its mass and power density, and must be greater than the maximum power required by the motor in a normal mission:

$$P_{max_m} < m_{mot} \cdot \rho_{mot} \tag{46}$$

(3) The maximum takeoff mass must exceed the total weight calculated in Section 3.2:

$$W_0 < mtom \cdot g \tag{47}$$

(4) The flight duration during the cruise phase is the longest, and the period requiring stable motor operation is also the longest. Therefore, the motor's rated power must exceed the power required during the cruise phase:

$$P_{cru_m} < P_{r_m} \tag{48}$$

(5) It is assumed that the aircraft needs to work 500 times per year and can work for 10 years. It is necessary to ensure that the capacity loss of the aircraft in this decade is less than 20 %, which means that the energy loss caused by the capacity loss is less than 20 %:

$$Q_{loss} \cdot U_r \cdot n_{cell} < 0.2 \cdot m_{bat} \cdot \rho_{bat} \tag{49}$$

where U_r is the rated voltage of the single cell during discharge, which is determined by the simulation results of MATLAB Simulink.

(6) The aircraft needs enough cruise velocity to ensure its smooth flight in the cruise phase:

$$\sqrt{\frac{2 \cdot W_0}{C_L \cdot \rho_2 \cdot S_{wi}}} < V \tag{50}$$

4.3. Optimization model

The total energy consumption model has been determined in Section 3.5, which is related to all design variables:

$$E_{total} = E_{total}(\mathbf{x}) \tag{51}$$

Incorporating the setup of various constraints, an optimization model for an eVTOL aircraft can be established with total energy consumption as the optimization objective:

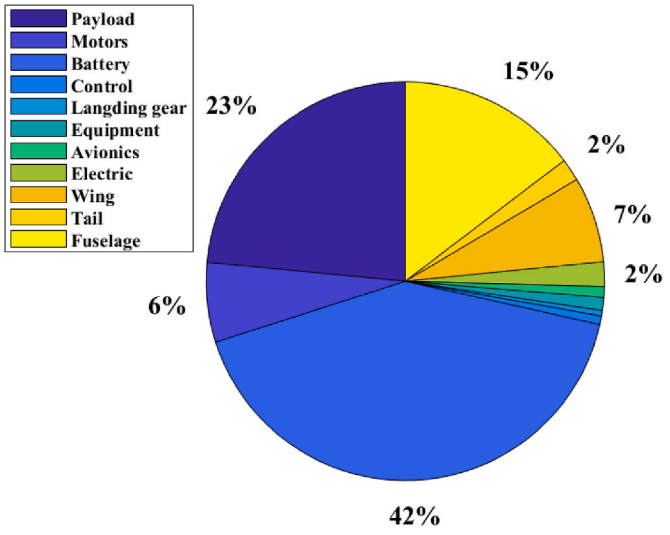


Fig. 14. Decomposition of maximum takeoff mass.

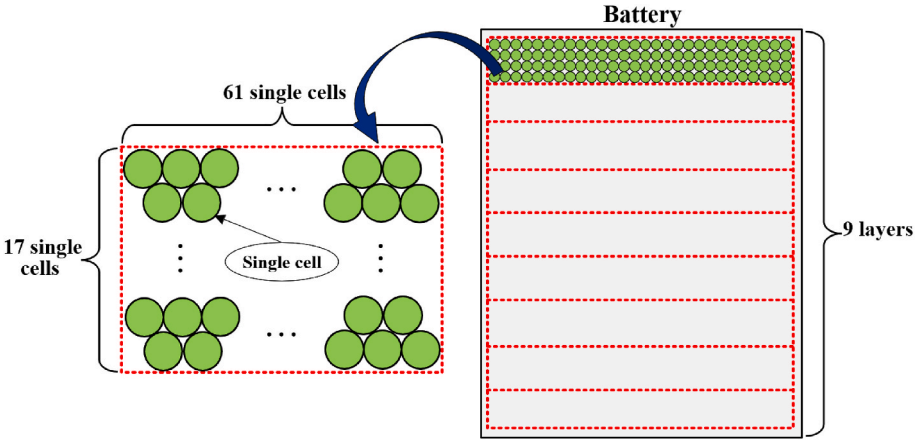


Fig. 13. Structure diagram of battery.

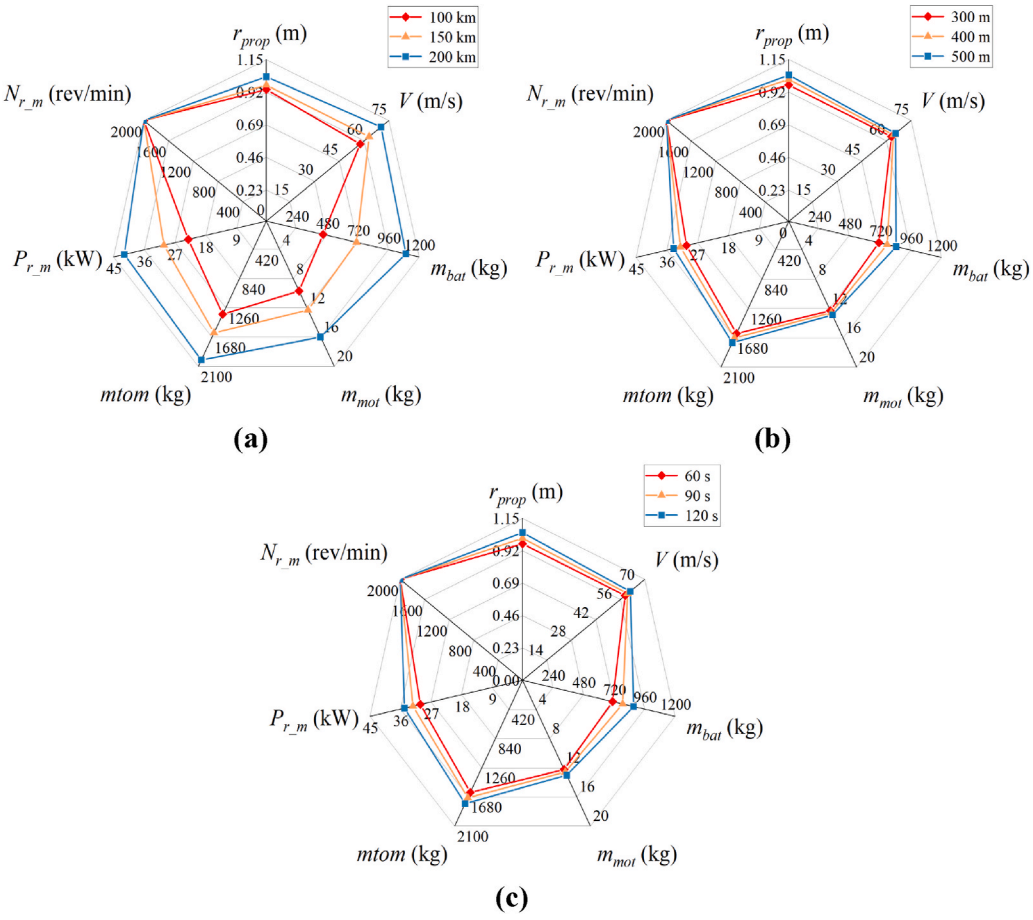


Fig. 15. The optimization results of design variables under different cruise ranges (a), heights (b), and hover times (c).

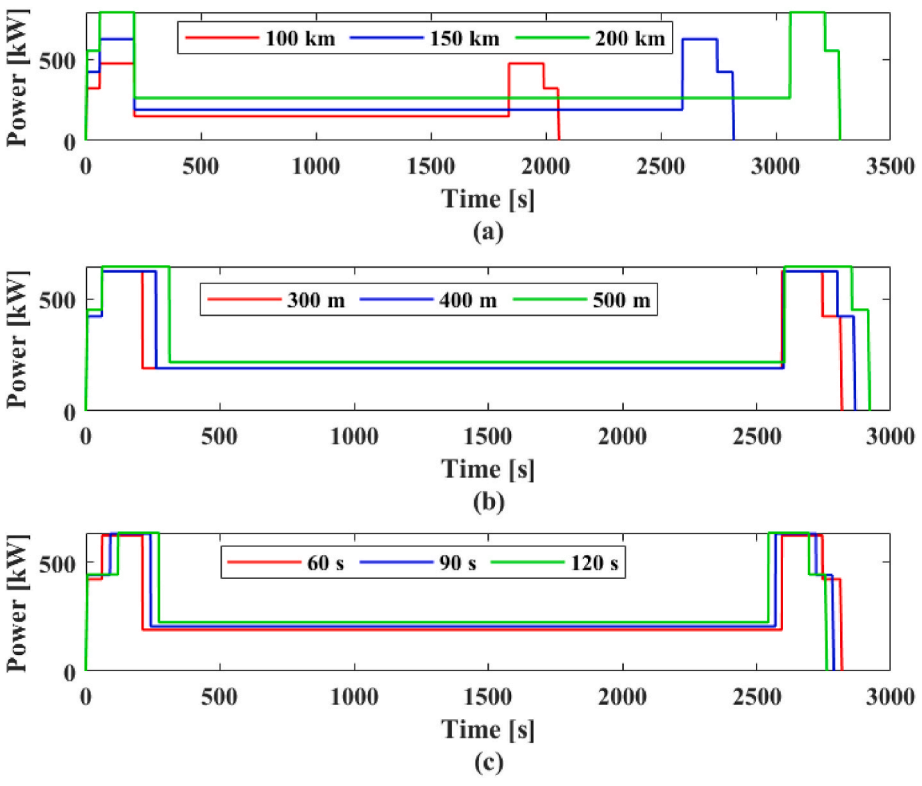


Fig. 16. The battery output power under different cruise ranges (a), heights (b), and hover times (c).

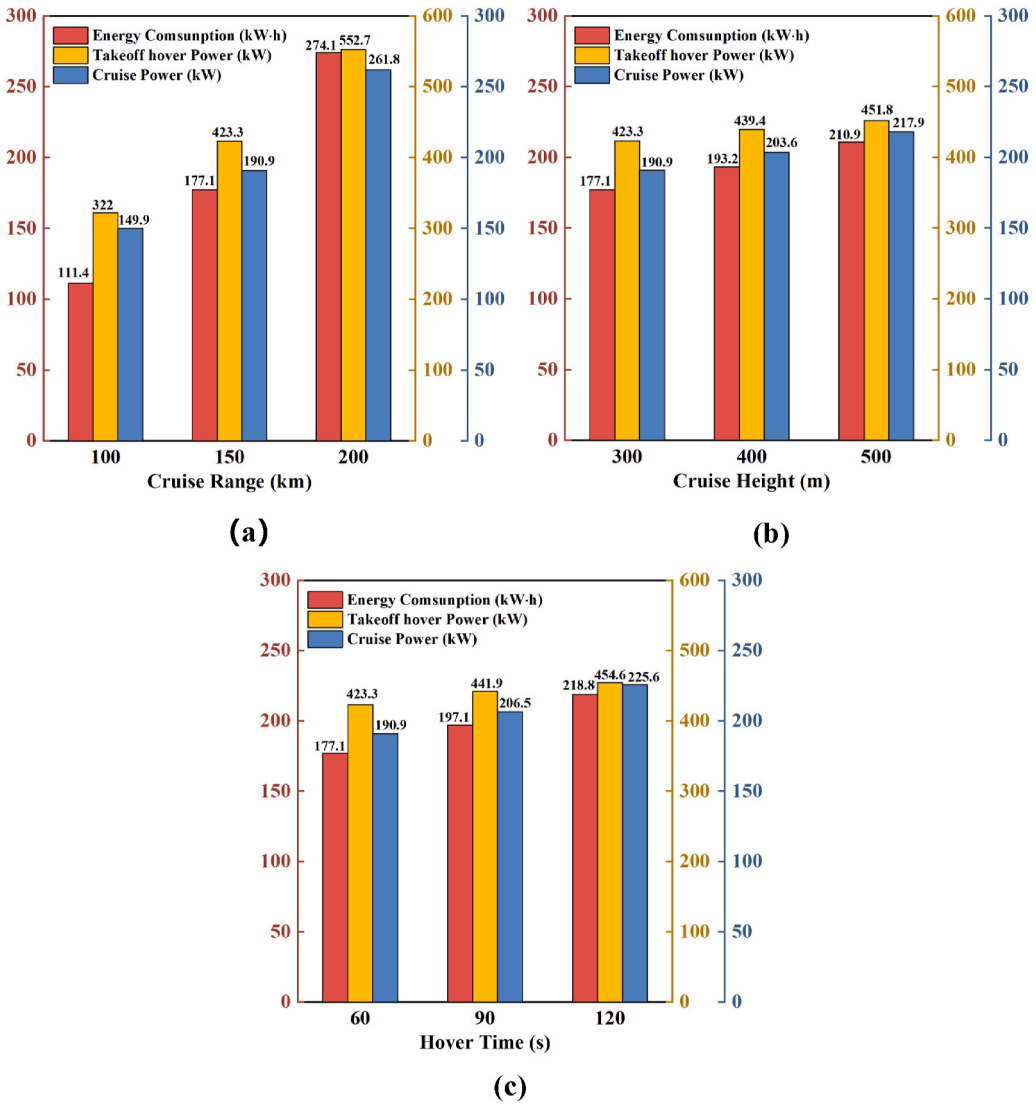


Fig. 17. The energy consumption as well as the battery power output during the takeoff hover and cruise phases for different cruise ranges (a), heights (b), and hover times (c).

$$\min\ E_{total}(\mathbf{x})$$

$$s.t.\begin{cases} \mathbf{x}_{min} \le \mathbf{x} \le \mathbf{x}_{max} \\ 1.2{\cdot}E_{total} - \dfrac{m_{bat}{\cdot}\rho_{bat}}{1000} < 0 \\ P_{max_m} - m_{mot}{\cdot}\rho_{mot} < 0 \\ W_0 - mtow{\cdot}g < 0 \\ P_{cru_m} - P_{r_m} < 0 \\ Q_{loss}{\cdot}U_r{\cdot}n_{cell} - 0.2{\cdot}m_{bat}{\cdot}\rho_{bat} < 0 \\ \sqrt{\dfrac{2{\cdot}W_0}{C_L{\cdot}\rho_2{\cdot}S_{wi}}} - V < 0 \end{cases} \tag{52}$$

The optimization framework was implemented using the well-established interior-point algorithm embedded in MATLAB's fmincon solver, augmented by a multidisciplinary feasible (MDF) coordination strategy to address interdisciplinary coupling in the design space. Nonlinear inequality constraints were explicitly defined within a dedicated nonlcon function, which was directly integrated into the fmincon workflow. Linear constraints, where applicable, were systematically specified through the standard matrix-vector interface using the A, b, Aeq, and beq parameters. Notably, the solver's native interior-point routine autonomously handles constraint feasibility via its integrated barrier function methodology, eliminating the need for auxiliary strategies such as external penalty methods or active-set adjustments. Gradient information for both the objective and constraint functions was numerically estimated by the solver's default finite-difference approximation scheme rather than analytically derived. This approach balances computational generality with implementation efficiency.

The optimization protocol was configured as follows.

(1) Maximum iterations: 5000 (to accommodate large-scale parameter exploration).
(2) Maximum function evaluations: 20,000 (ensuring exhaustive search within feasible regions).
(3) Numerical Differentiation: Central difference scheme (selected over forward differences for improved gradient accuracy).
(4) Step tolerance: 1e−6 (halting iterations when parameter updates fell below this threshold).
(5) Constraint violation tolerance: 1e−6 (enforcing strict adherence to feasibility boundaries).

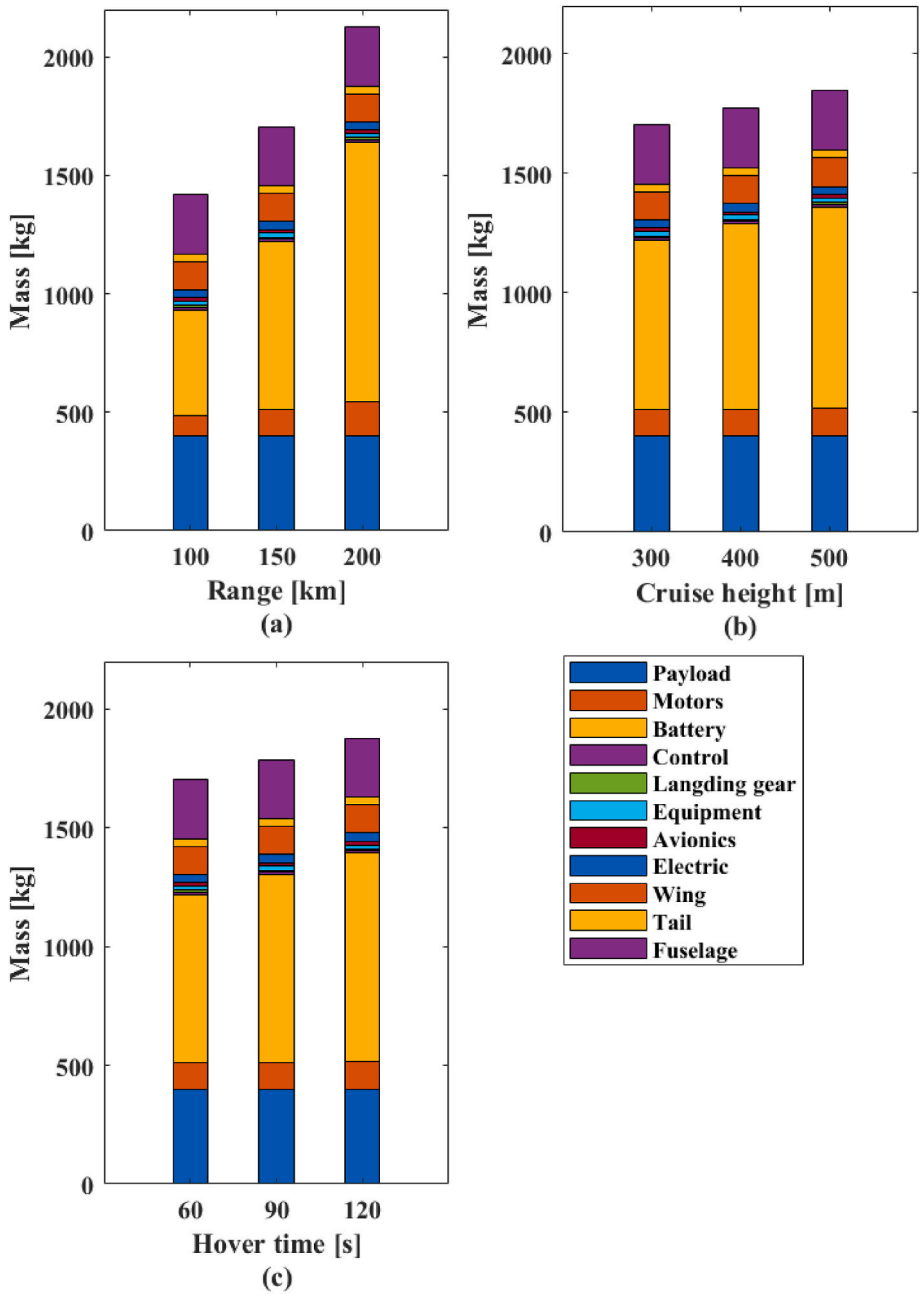


Fig. 18. The decomposition of maximum takeoff mass under different cruise ranges (a), heights (b), and hover times (c).

4.4. Optimization results and discussion

4.4.1. Comparison of different design schemes

The optimization results were compared with an existing study [48] to highlight the strengths of this work, with that study serving as the baseline. Before the comparison, the baseline parameters were adjusted to ensure consistency with the current research as follows.

(1) Since the baseline did not account for ESC efficiency, the ESC efficiency of the baseline was set to 0.85, consistent with the parameters used in this paper.
(2) The baseline only considered energy consumption during the cruise and takeoff hover phases. To align with this study, it was assumed that the energy consumption of the control group during the climb, descent, and landing hover phases is 2.8 times that of the takeoff hover phase.
(3) The baseline was assumed to have the same cruise range (150 km) and payload (400 kg) as the aircraft in this study.
(4) The baseline was also assumed to have the same initial values and value ranges for design variables as in this research.

A comparison of the design parameters optimization results is shown in Table 6.

The aircraft designed by the parameters presented in this study exhibits a total energy consumption of 177.13 kW h, representing an 11.44 % reduction compared to the baseline value of 200 kW h. Additionally, the maximum takeoff mass of the aircraft is decreased by 15.81 % relative to the baseline, contributing to a more lightweight design.

From the final optimization results, the required number of single cells for the battery is determined to be 9322.5, which is rounded up to 9333. Considering the dimensions of the battery and the aircraft (length, width, and height), the final configuration of the single cells is arranged

as follows: 9333 = 61 × 17 × 9 (61 single cells connected in series, 17 single cells connected in parallel, with this configuration repeated in 9 layers). The battery structure of the aircraft is illustrated in Fig. 13.

The maximum takeoff mass of the aircraft is broken down as shown in Fig. 14. It is found that the mass of the battery is almost close to half of the maximum takeoff mass, which is one of the reasons why electric manned aircraft are difficult to popularize in modern times. The low energy density of the batteries leads to the need to pair them with a large weight of batteries to drive a large tonnage aircraft to take off. In the event of a rapid proliferation of all-electric aircraft, the primary technological consideration would be the enhancement of battery energy density.

4.4.2. Comparison of different missions

This section evaluates the effects of different cruise ranges (100 km, 150 km, 200 km), cruise heights (300 m, 400 m, 500 m), and hover times (60 s, 90 s, 120 s) on optimization results while maintaining constant other parameters. Fig. 15 illustrates the optimization results of design variables under these different flight mission profiles.

Figs. 15 and 16 illustrate the battery output power under different cruise ranges, heights, and hover times. Additionally, Fig. 17 shows the energy consumption as well as the battery power output during the takeoff hover and cruise phases for different missions. The energy consumption and battery output power increase with an increase in cruise range, height, and hover time. Furthermore, although the time of the takeoff hover phase is shorter than that of the cruise phase, the battery power consumption is comparable between the two phases. Therefore, future research should focus on reducing energy consumption during the takeoff hover phase.

The decomposition of maximum takeoff mass under different flight mission profiles is illustrated in Fig. 18. Analysis of the data in the figure reveals that apart from the motor and battery, the mass of the aircraft's other components remains largely constant. This phenomenon occurs because the aircraft's geometric configuration was predefined before the optimization design phase, which is not ideal in practical aircraft design processes. However, since energy consumption primarily originates from the propulsion system, variations in the masses of other components are not the focus of this study. Due to constraints imposed by the maximum power demand, the mass of the motor has increased, yet the increment is relatively small. Simultaneously, the battery mass continues to increase, with a comparatively larger increment, as it is constrained by battery energy density. This indicates that battery mass is significantly influenced by factors such as cruise range, height, and hover time, necessitating thorough consideration during the design process.

Through the analysis of energy consumption, power requirements, and mass distribution of the aircraft under various flight missions, it is evident that flight mission parameters significantly impact the aircraft's performance and design features. Consequently, in the aircraft design process, it is crucial to comprehensively consider the intended flight missions, ensuring that all design aspects align with the specified operational requirements.

5. Conclusion

This study addresses the energy consumption optimization of eVTOL aircraft and proposes a methodology based on MDO. The approach comprehensively considers weight, motor efficiency, and the electrochemical-aging-thermal coupled mechanisms of the battery, establishing a multi-disciplinary coupled model for aircraft energy consumption. The findings reveal that these coupling effects significantly influence the energy consumption and performance of the aircraft. By developing a systematic MDO model and applying optimization algorithms, the study identifies optimal design parameters across various flight missions.

Study results demonstrate that the proposed optimization scheme outperforms the baseline design in key performance metrics: achieving an 11.44 % reduction in total energy consumption and a 15.81 % decrease in maximum takeoff mass. Furthermore, a comparative analysis of optimized results for different flight missions shows that variables such as cruise range, height, and hover time significantly impact battery weight, and the performance of the battery is identified as the critical factor in determining the overall efficiency of eVTOL aircraft.

Despite the progress made in energy consumption optimization for eVTOL aircraft, the study acknowledges several limitations: first, due to the complexity of the model, certain variables were subject to idealized assumptions, failing to fully account for dynamic changes and uncertainties; second, due to data limitations, the study utilized A123 $LiFePO_4$ cylindrical batteries, which may not match the characteristics required for aviation-specific applications. Future research necessitates the incorporation of uncertainty considerations into a multi-disciplinary framework, integrating additional disciplines to enhance the methodology. By combining global sensitivity analysis with multi-level hierarchical coordination algorithms, robust design optimization for eVTOL will be investigated. Additionally, there is a need to prioritize the development of novel aerospace battery designs.

CRediT authorship contribution statement

Wei Li: Writing – review & editing, Validation, Supervision, Methodology. **Rongguo Cheng:** Writing – original draft, Software, Methodology, Data curation. **Haihong Huang:** Writing – review & editing, Investigation, Formal analysis, Conceptualization. **Akhil Garg:** Writing – review & editing, Methodology, Formal analysis, Data curation. **Liang Gao:** Writing – review & editing, Supervision.

Declaration of competing interest

We declare that we do not have any commercial or associative interest that represents a conflict of interest in connection with the work submitted.

Acknowledgment

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Appendix

The calculation formulas for each component are presented in Ref. [27].

W_{eq} is calculated from Eq. (A.1).

$$W_{eq} = 0.0582 \cdot W_0 - 65 \tag{A.1}$$

The structural weight consists of wing weight W_{wi}, tail weight W_{ta}, and fuselage weight W_{fu}:

$$W_{\mathrm{str}} = W_{wi} + W_{ta} + W_{fu} \tag{A.2}$$

The wing weight is calculated by Eq. (A.3):

$$W_{wi} = 0.036 \cdot S_{wi}^{0.76} \cdot \left(\frac{AR_w}{\cos^2 \Lambda_{c/4}}\right)^{0.6} \cdot q^{0.006} \cdot \lambda_{wi}^{0.04} \cdot \left(\frac{100 \cdot \frac{t}{c}}{\cos^2 \Lambda_{c/4}}\right)^{-0.3} \cdot (n_z \cdot W_0)^{0.49} \tag{A.3}$$

where S_{wi} is the reference area of the wing. AR_w is the wing aspect ratio. $\Lambda_{c/4}$ is the sweep angle at the quarter chord of the wing. q is the dynamic pressure during cruising. λ_{wi} is the wing tip-root ratio. t/c is the relative thickness of the wing. n_z is the maximum overload, and 6 g is taken here.

The tail weight is divided into horizontal tail weight W_{hT} and vertical tail weight W_{vT}, which are calculated by Eq. (A.4) and Eq. (A.5) respectively:

$$W_{hT} = 0.016 \cdot S_{hT}^{0.896} \cdot \left(\frac{AR_{hT}}{\cos^2 \Lambda_{hT}}\right)^{0.6} \cdot q^{0.168} \cdot \lambda_{hT}^{-0.12} \cdot \left(\frac{100 \cdot \frac{t_{hT}}{c_{hT}}}{\cos^2 \Lambda_{hT}}\right)^{-0.12} \cdot (n_z \cdot W_0)^{0.414} \tag{A.4}$$

where S_{hT} is the reference area of the horizontal tail; AR_{hT} is the aspect ratio of the horizontal tail. Λ_{hT} is the sweepback angle at the chord of the horizontal tail. λ_{hT} is the tip-root ratio of the horizontal tail. t_{hT}/c_{hT} is the relative thickness of the horizontal tail.

$$W_{vT} = 0.073 \cdot S_{vT}^{0.87} \cdot \left(\frac{AR_{vT}}{\cos^2 \Lambda_{vT}}\right)^{0.36} \cdot q^{0.122} \cdot \lambda_{vT}^{-0.03} \cdot \left(\frac{100 \cdot \frac{t_{vT}}{c_{vT}}}{\cos \Lambda_{hT}}\right)^{-0.49} \cdot (n_z \cdot W_0)^{0.376} \tag{A.5}$$

where S_{vT} is the reference area of the vertical tail. AR_{vT} is the aspect ratio of the vertical tail. Λ_{vT} is the sweepback angle at the chord of the vertical tail. λ_{vT} is the tip-root ratio of the vertical tail. t_{vT}/c_{vT} is the relative thickness of the vertical tail.

The fuselage weight is calculated by Eq. (A.6):

$$W_{fu} = 0.052 \cdot S_{fu}^{1.086} \cdot (n_z \cdot W_0)^{0.18} \cdot l_{hT}^{-0.05} \cdot \left(\frac{l_{fs}}{d_{fs}}\right)^{-0.07} q^{0.241} + 11.9 \cdot (V_P \cdot \Delta P)^{0.27} \tag{A.6}$$

where, S_{fu} is the surface area of the fuselage, which is calculated by Eq. (A.7). l_{hT} is the horizontal tail arm. l_{fs} is the length of the fuselage structure. d_{fs} is the height of the fuselage structure. V_P is the cabin volume. ΔP is the cabin pressure difference, and 8 psi is taken here.

$$S_{fu} = \frac{\pi \cdot D}{4} \cdot \left\{ \frac{1}{3 \cdot L_1^2} \cdot \left[\left(4 \cdot L_1^2 + \frac{D^2}{4}\right)^{1.5} - \frac{D^3}{8} \right] - D + 4 \cdot L_2 + \frac{2 \cdot (D + d_1)}{D} \cdot \sqrt{L_3^2 + \frac{D^2 - d_1^2}{4}} + \frac{2 \cdot (d_1 + d_2)}{D} \cdot \sqrt{L_4^2 + \frac{d_1^2 - d_2^2}{4}} \right\} \tag{A.7}$$

The parameters in equation (A.7) are given in Fig. A.1.

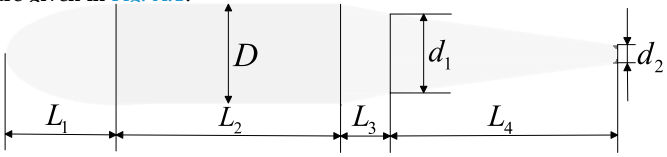


Fig. A.1. Geometry data of fuselage

The control system weight is calculated by Eq. (A.8):

$$W_{con} = 0.053 \cdot l_{fs}^{1.54} \cdot b^{0.37} \cdot (n_z \cdot W_0 \times 10^{-4})^{0.8} \tag{A.8}$$

where b is the wing span.

The landing gear weight is calculated by Eq. (A.9):

$$W_{lg} = 0.054 \cdot (n_z \cdot W_0)^{0.68} \cdot \left(\frac{l_l}{12}\right)^{0.6} \tag{A.9}$$

where l_l is the length of landing gear, and 1.23 m is taken here.

The avionics weight is set at 15 × 9.8 N.

The electric weight is calculated by Eq. (A.10):

$$W_{el} = 12.57 \cdot (W_{av})^{0.51} \tag{A.10}$$

The values of the above geometric parameters obtained from Fig. 3 are shown in Table A.1.

Table A.1
Geometric parameters

Component	Parameter	Value
Wing	S_{wi}	13.04 m^2
	AR_w	10.12
	$\Lambda_{c/4}$	0°
	λ_{wi}	0.53
	t/c	0.15
	b	11.49 m
Horizontal tail	S_{hT}	5.06 m^2
	AR_{hT}	2.95
	Λ_{hT}	5°
	λ_{hT}	0.66
	t_{hT}/c_{hT}	0.1
	l_{hT}	13.5
Vertical tail	S_{vT}	1.47 m^2
	AR_{vT}	1.73
	Λ_{vT}	18°
	λ_{vT}	0.5
	t_{vT}/c_{vT}	0.1
Fuselage	l_{fs}	6.73 m
	d_{fs}	2.30 m
	V_P	5.19 m^3
	D	2.47 m
	d_1	0.72 m
	d_2	0.21 m
	L_1	1.65 m
	L_2	2.47 m
	L_3	0.58 m
	L_4	2.03 m

Data availability

Data will be made available on request.

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